SAVEETHA SCHOOL OF ENGINEERING

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17.

Aim: To write a Prolog program to calculate the sum of the first 'n' integers.

Algorithm:

- **Step 1:** Define the base case fact where the sum of 0 is 0 (sum(0,0).).
- **Step 2:** Define the recursive rule sum(N,S) that applies when the integer N is greater than 0.
- **Step 3:** Calculate the previous integer N1 by subtracting 1 from N (N1 is N - 1).
- **Step 4:** Recursively call the sum predicate using N1 to find its corresponding sum S1 (sum(N1,S1)).
- **Step 5:** Calculate the final sum S by adding N to S1 (S is S1 + N).

PROGRAM:

```
/* sum of n integers */


sum(0,0).


sum(N,S) :-
        N > 0,
        N1 is N -1,
        sum(N1,S1),
        S is S1 + N.
```

OUTPUT:

Result: Thus, the program was executed and found that the output was found to be correct.

18.BIRTHDAY

Aim:

To write a Prolog program to establish a database of birthdays and execute queries regarding birth years, birth months, and age comparisons.

Algorithm:

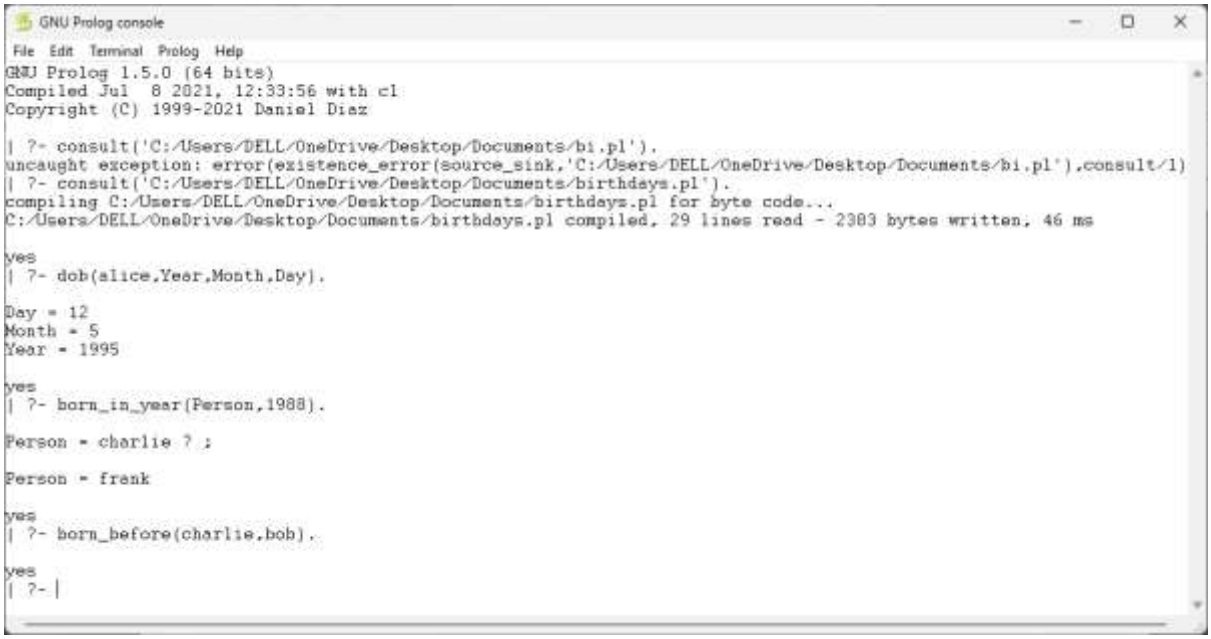
- **Step 1:** Define the database facts specifying the date of birth for various individuals using the format dob(Name, Year, Month, Day).
- **Step 2:** Create Rule 1 (born_in_year) to find individuals born in a specific year by extracting the Year variable from the dob fact.
- **Step 3:** Create Rule 2 (born_in_month) to find individuals born in a specific month by extracting the Month variable from the dob fact.
- **Step 4:** Create Rule 3 (born_before) to compare the birth years of two distinct people to check if the first person's birth year is less than the second person's.
- **Step 5:** Create Rule 4 (same_birth_year) to identify pairs of different individuals who share the exact same birth year.

PROGRAM:

```
% FACTS: dob(Name, Year, Month, Day)


dob(alice, 1995, 5, 12).

dob(bob, 2001, 11, 23).

dob(charlie, 1988, 5, 5).

dob(david, 1995, 8, 30).

dob(eve, 2005, 2, 14).

dob(frank, 1988, 12, 1).


% RULES

% Rule 1: Find people born in a specific year

born_in_year(Name, Year) :-
```

```
    dob(Name, Year, _, _).
% Rule 2: Find people born in a specific month (1 = Jan, 2 = Feb, etc.)
born_in_month(Name, Month) :-
    dob(Name, _, Month, _).
% Rule 3: Check if Person1 was born in an earlier year than Person2
born_before(Person1, Person2) :-
    dob(Person1, Year1, _, _),
    dob(Person2, Year2, _, _),
    Year1 < Year2.
% Rule 4: Find pairs of people who were born in the same year
same_birth_year(Person1, Person2) :-
    dob(Person1, Year, _, _),
    dob(Person2, Year, _, _),
    Person1 \= Person2.
```

OUTPUT:



Result: Thus, the program was executed and found that the output was found to be correct.

19.STUDENT TEACHER DATABASE

Aim:

To write a Prolog program that models a student-teacher relational database to extract information about course enrollments, instructors, and classmates.

Algorithm:

- Step 1: Define facts mapping teachers to the specific subject codes they teach (teaches(Teacher, SubjectCode)).
- Step 2: Define facts mapping students to the specific subject codes they are enrolled in (enrolled(Student, SubjectCode)).
- Step 3: Define facts associating subject codes with their full subject names (subject(SubjectCode, SubjectName)).
- Step 4: Create Rule 1 (teacher_of) to establish that a teacher teaches a student if they both share the same subject code.
- Step 5: Create Rule 2 (student_in_class) to retrieve students taught by a specific teacher for a specific subject code.
- Step 6: Create Rule 3 (classmates_by_teacher) to verify if two distinct students are taught by the same teacher.
- Step 7: Create Rule 4 (student_course_details) to consolidate full course information, retrieving the student, subject code, subject name, and teacher simultaneously.

PROGRAM:

```
% teaches(Teacher, SubjectCode)

teaches(dr_smith, cs101).

teaches(dr_smith, cs102).

teaches(prof_jones, math201).

teaches(prof_davis, phys101).

teaches(dr_wilson, cs101).
```

```
% enrolled(Student, SubjectCode)
enrolled(alice, cs101).
enrolled(alice, math201).
enrolled(bob, cs102).
enrolled(charlie, phys101).
enrolled(charlie, cs101).
enrolled(david, math201).

% subject(SubjectCode, SubjectName)
subject(cs101, 'Introduction to Computer Science').
subject(cs102, 'Data Structures').
subject(math201, 'Calculus II').
subject(phys101, 'General Physics I').



% Rule 1: A teacher teaches a student if they teach a subject code the student is enrolled in.
teacher_of(Teacher, Student) :-
    teaches(Teacher, SubjectCode),
    enrolled(Student, SubjectCode).

% Rule 2: Find all students taught by a specific teacher for a specific subject code.
student_in_class(Student, Teacher, SubjectCode) :-
    teaches(Teacher, SubjectCode),
    enrolled(Student, SubjectCode).

% Rule 3: Check if two students share the same teacher.
```

```
classmates_by_teacher(Student1, Student2, Teacher) :-
    teacher_of(Teacher, Student1),
    teacher_of(Teacher, Student2),
    Student1 \= Student2.

% Rule 4: Get full course info for a student (Student, SubjectCode, SubjectName, Teacher)
student_course_details(Student, SubjectCode, SubjectName, Teacher) :-
    enrolled(Student, SubjectCode),
    subject(SubjectCode, SubjectName),
    teaches(Teacher, SubjectCode).
```

OUTPUT:

Result: Thus, the program was executed and found that the output was found to be correct.

20.PLANETS DB

Aim:

To write a Prolog program to structure a database of planetary characteristics and query them based on attributes like rings, moons, distance, and classification.

Algorithm:

- Step 1: Define facts for various planets detailing their name, classification type, distance from the sun in millions of kilometers, number of moons, and presence of rings (planet_info).
- Step 2: Create Rule 1 (has_rings) to filter and find planets where the ring attribute is listed as 'yes'.
- Step 3: Create Rule 2 (has_moons) to filter and find planets where the recorded number of moons is strictly greater than 0.
- Step 4: Create Rule 3 (planet_type) to query and categorize planets by their specific classification type (e.g., terrestrial, gas_giant).
- Step 5: Create Rule 4 (closer_to_sun) to compare the distances of two given planets and determine if the first has a smaller distance than the second.
- Step 6: Create Rule 5 (many_moons) to identify heavily moon-rich planets by checking if their moon count exceeds 10.

PROGRAM:

```
% FACTS: planet_info(Name, Type, DistanceFromSun_Million_Km, Moons, HasRings)

planet_info(mercury, terrestrial, 58, 0, no).
planet_info(venus, terrestrial, 108, 0, no).
planet_info(earth, terrestrial, 150, 1, no).
planet_info(mars, terrestrial, 228, 2, no).
planet_info(jupiter, gas_giant, 778, 95, yes).
planet_info(saturn, gas_giant, 1427, 146, yes).
planet_info(uranus, ice_giant, 2871, 27, yes).
planet_info(neptune, ice_giant, 4497, 14, yes).

% Rule 1: Find planets that have rings
has_rings(Planet) :-
    planet_info(Planet, _, _, _, yes).

% Rule 2: Find planets that have at least one moon
has_moons(Planet) :-
    planet_info(Planet, _, _, Moons, _),
    Moons > 0.

% Rule 3: Find planets by their classification type (e.g., terrestrial, gas_giant)
planet_type(Planet, Type) :-
    planet_info(Planet, Type, _, _, _).

% Rule 4: Check if Planet1 is closer to the sun than Planet2
```

```
closer_to_sun(Planet1, Planet2) :-
    planet_info(Planet1, _, Dist1, _, _),
    planet_info(Planet2, _, Dist2, _, _),
    Dist1 < Dist2.

% Rule 5: Find heavily heavily cratered/moon-rich planets (more than 10 moons)
many_moons(Planet) :-
    planet_info(Planet, _, _, Moons, _),
    Moons > 10.
```

OUTPUT:

Result: Thus, the program was executed and found that the output was found to be correct.